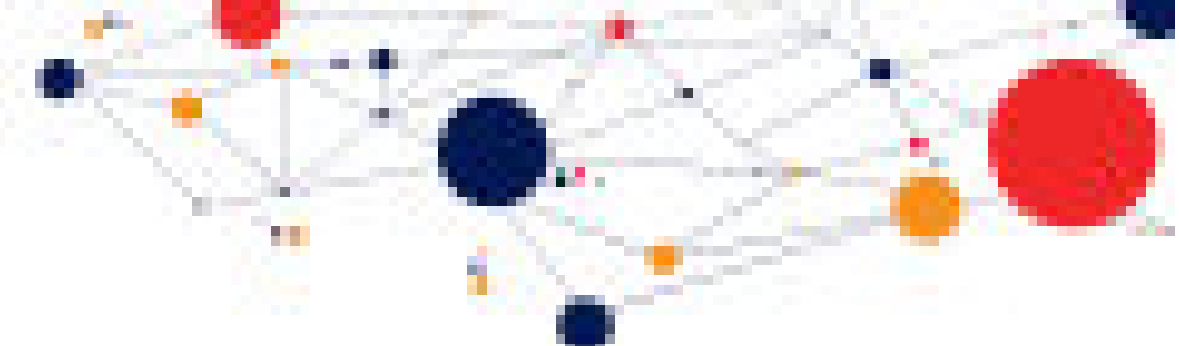
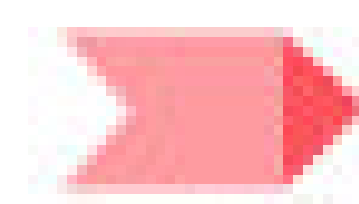
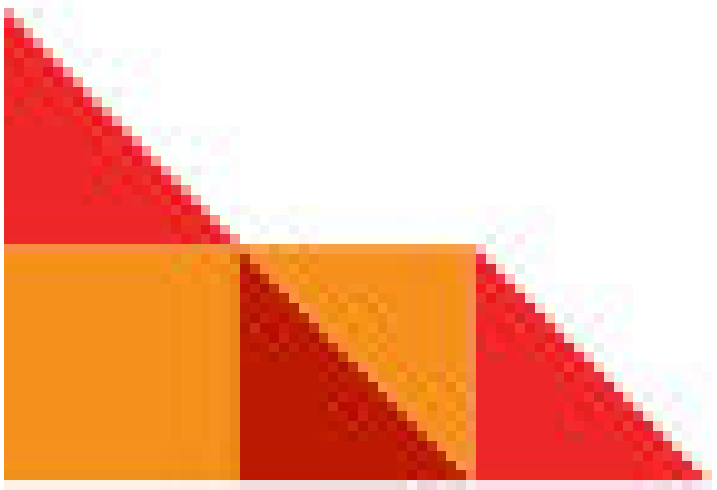


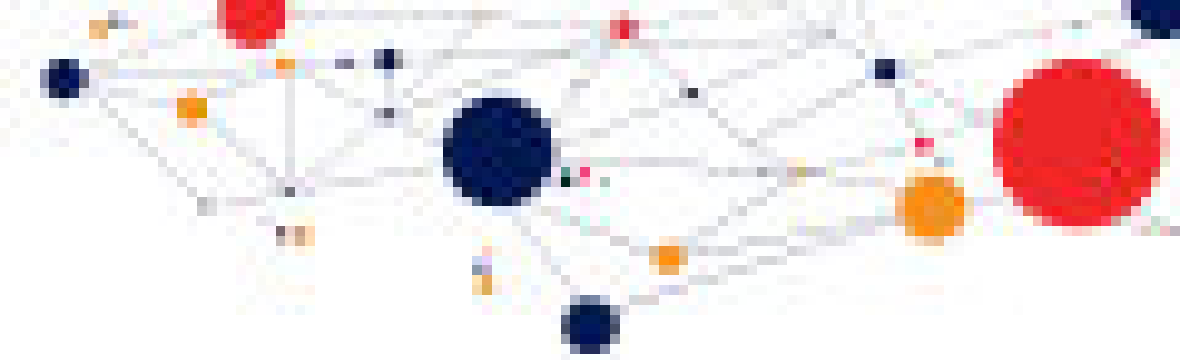
BSE INDEX SERVICES PVT. LTD.

***Automation & Analytical Enhancement of Free Float Factor Identification
& Creation of BSE Sector Rotation Index (BSRI)***



Product Manager and Data Analyst Intern





Earlier Methodology at BSE to Identify FreeFloat Shares

Refer previous quarter FFF values; select stocks to check, if difference is > 2%

Visit BSE Website for each selected stock (800+ per quarter, 3 min / stock)

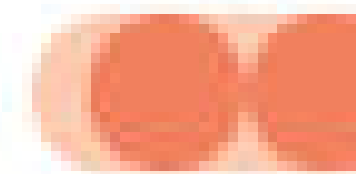
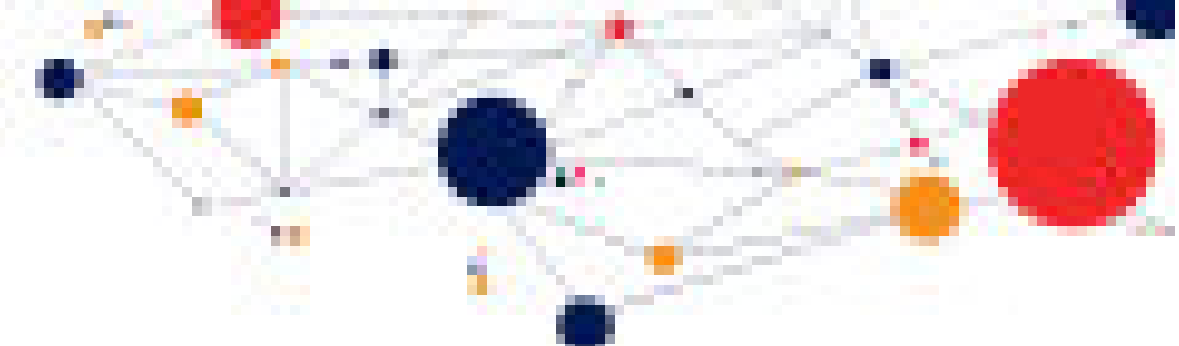
Visit annual reports for exceptional cases for analysis (50 cases, 10 min/stock)

Mark each NEW shareholder as Strategic, Promotor OR Public Shareholders

Calculate new free float values by inputting the changes and running formulas

Free Float Shares?

Root Cause of Partial Structured Shareholding Pattern : Many Companies try to maximize their Free Float Shares by declaring Strategic Promoters Shareholders as Public



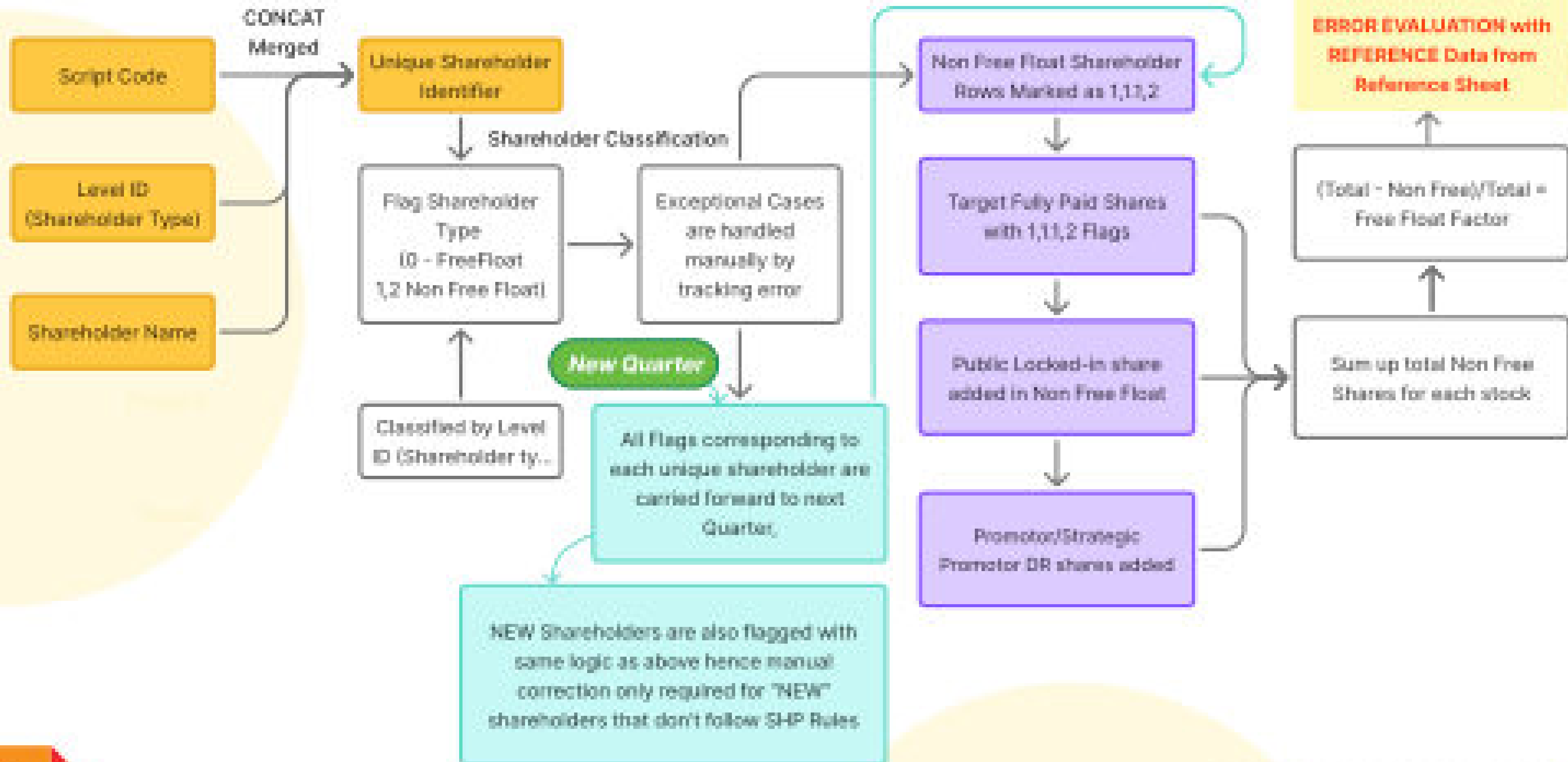
Objectives (Free Float Factor Model)

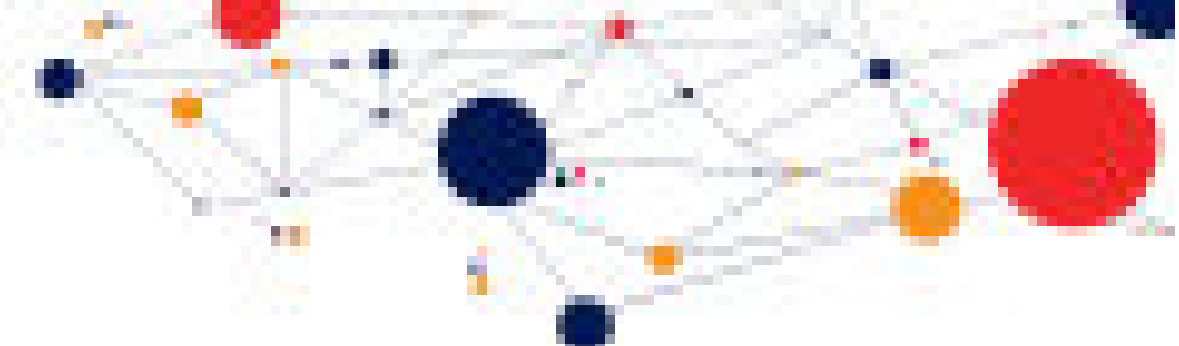
- Standardize Free Float Factor calculations across BSE indices using automation
- Reduce operational time and manual errors in index management and reporting

Methodology (Free Float Factor Model)

- Excel-SQL model to automate shareholder classification and Free Float Factor (FFF) computation
- Rule-based flags for Promoter, Strategic, Doubtful, and Public shareholders
- Quarterly data validation against historical records to ensure compliance and accuracy

Methodology (Free Float Factor Model)

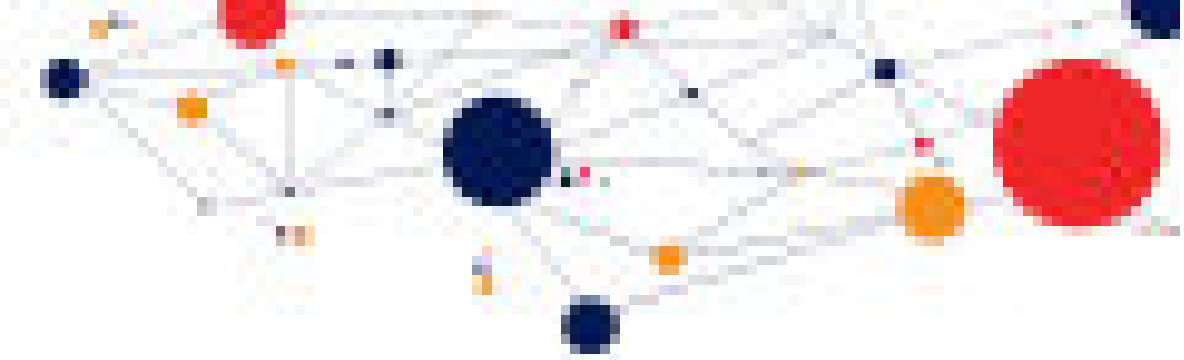




Results & Impact (FFF Model)

- ***Boosted daily BSE trade volume by ₹700 Cr (0.7%)***
- ***Reduced process time: 20 days/quarter x 3 FTEs – 1 day/quarter x 1 FTE (80x faster)***
- ***Eliminated 1–5% error in Free Float shares for 300 stocks out of 4,800+***
- ***Increased Free Float Shares Weightage in every BSE Index (Even Sensex)***
- ***Enabled accurate root-cause tracing and compliance with global standards***

PROJECT 2 : BSE SECTOR ROTATION INDEX



Pull & download all Index data (Date, Values, Constituents, etc) internally

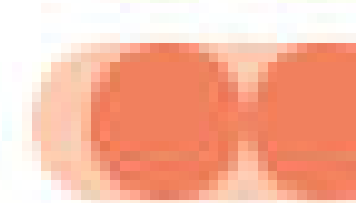
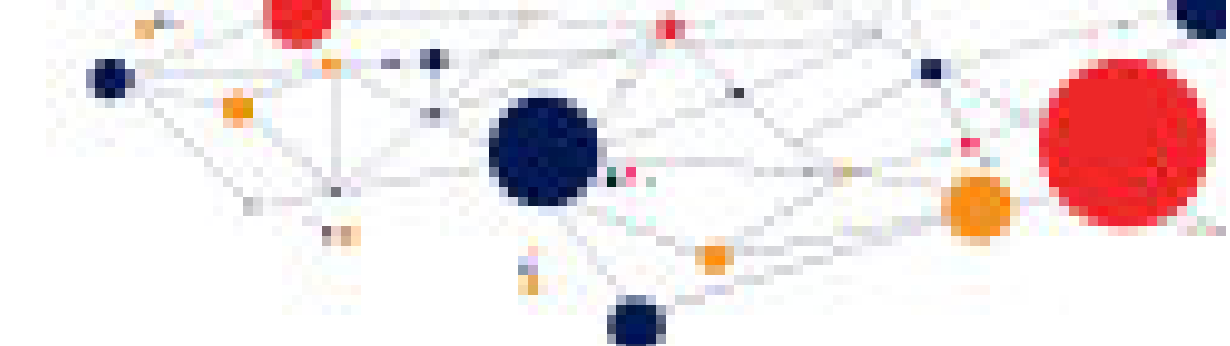
Pull & download all Index stocks constituent data (Sector, Date, Price, Weight)

Copy Paste both data in a Standard Excel formula template

Calculate Index Performances, Index details for each month for 101 Index

Create Sales PPT on MS Powerpoint before meeting Wealth Management Firms

Organize all 101 indices every month, and clear them next month



Objectives (Sales Deck Automation)

- Develop automated generation of branded, analytics-rich client presentations
- Reduce manual reporting workload for Sales team
- Enhance client delivery speed and consistency across 101 indices

Methodology (Sales Deck Automation)

- Decide what data needs to be shown to client
- Build presentation design using Figma
- Collaborate with Front-End Team to build the Presentation ready for PDF conversion
- Collaborate with Data team to arrange required raw data
- Collaborate with Python & SQL team utilizing raw data for final calculations
- Collaborate with Backend team to bind final data with frontend and make the PDFs downloadable on a secure web portal just by selecting index type and Month

BSE MIDCAP

ASX MIDCAP PER. 2021

Overview

30 Sep 2021

The BSE MIDCAP provides real-time performance data. It is designed to measure the performance of the 50 largest, non-listed mid-cap companies relative to the ASX MIDCAP index. The index is calculated as the sum of the 50 largest, non-listed mid-cap companies relative to the ASX MIDCAP index.

Key Facts

Indexable	10 January 2021
Indexable Date	10 January 2021
Base Price	100
Number of Constituents	10
Indexable Frequency	Quarterly (Jan, Apr, Jul, Oct)
Weighting Method	Free float market cap weighted
Index Return	100%
Indexable Constituents	100%

Systematic

100	100
100	100
Standard Deviation	100

Top Constituents by Weightage

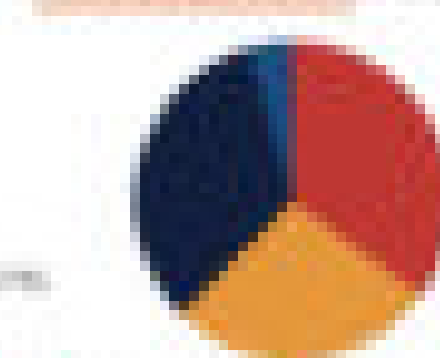
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%
100%	100%

Expected Composition of BSE MIDCAP



- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%

Market Cap Size Coverage



- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%
- 100%

BSE MIDCAP

ASX MIDCAP PER. 2021

Return

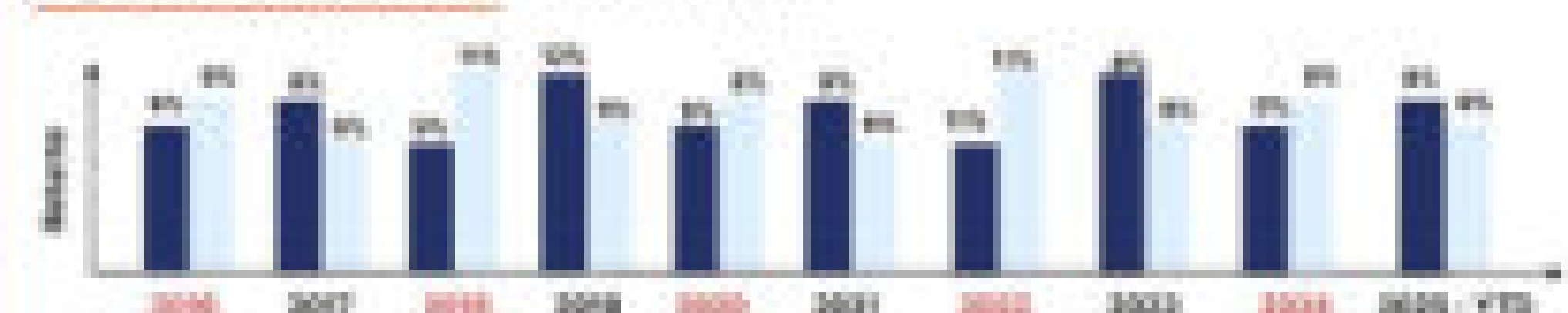
30 Sep 2021

Indexable	1 Month	3 Month	YTD	1 Year	3 Years	5 Years	10 Years
BSE MIDCAP							
ASX MIDCAP							

Real Return

Indexable	1 Month	3 Month	YTD	1 Year	3 Years	5 Years	10 Years
BSE MIDCAP							
ASX MIDCAP							

Comparison of BSE MIDCAP Calendar Year Returns with ASX MIDCAP



BSE MIDCAP

ASX MIDCAP PER. 2021

Market Returns

30 Sep 2021

	1 Year	3 Years	5 Years	7 Years	10 Years
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					

BSE MIDCAP

ASX MIDCAP PER. 2021

Performance

30 Sep 2021

Performance during Market Cycle

Index	1 Year	3 Years	5 Years	7 Years	10 Years
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					

Performance during Market Cycle

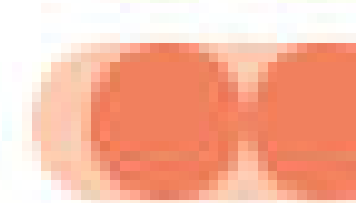
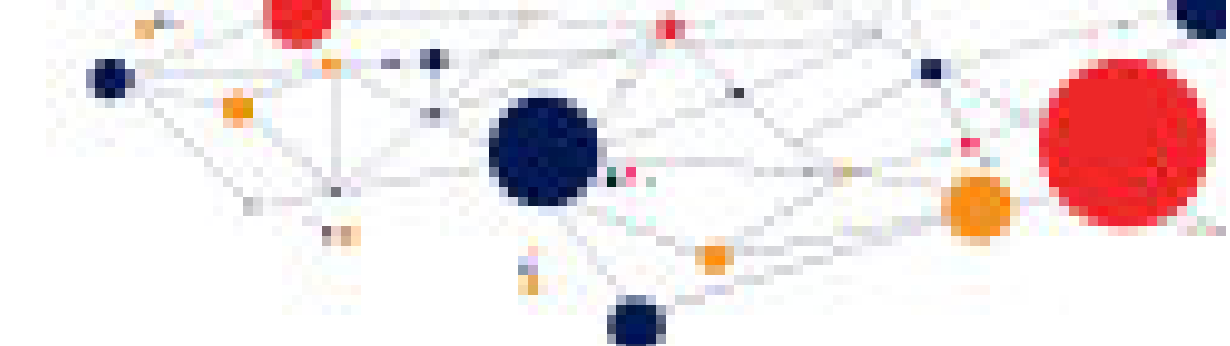
Index	1 Year	3 Years	5 Years	7 Years	10 Years
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					

Index 10 performance (BSE MIDCAP)

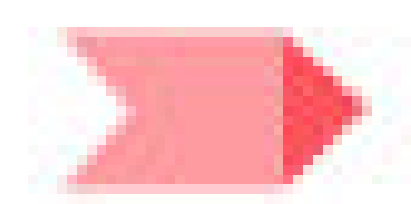
Index	1 Year	3 Years	5 Years	7 Years	10 Years
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					

Index 10 performance (ASX MIDCAP)

Index	1 Year	3 Years	5 Years	7 Years	10 Years
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					
BSE MIDCAP					
ASX MIDCAP					



Results (Sales Deck Automation)

- *90% reduction in manual prep; saved 30+ hrs/month for Sales team*
 - *Client turnaround time improved by 20%*
 - *Consistent, error-free, branded presentations now standard at BSE*
 - *Portal adopted as sole tool for sales reporting across all indices*
- 

THANK YOU 

